

Filter Bank Regularized Common Spatial Pattern Ensemble for Small Sample Motor Imagery Classification

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Abstract—For the last few years, many feature extraction methods have been proposed based on biological signals. Among these, the brain signals have the advantage that they can be obtained, even by people with peripheral nervous system damage. Motor imagery electroencephalograms (EEG) are inexpensive to measure, offer a high temporal resolution, and are intuitive. Therefore, these have received a significant amount of attention in various fields, including signal processing, cognitive science, and medicine. The common spatial pattern (CSP) algorithm is a useful method for feature extraction from motor imagery EEG. However, performance degradation occurs in a small-sample setting (SSS), because the CSP depends on sample-based covariance. Since the active frequency range is different for each subject, it is also inconvenient to set the frequency range to be different every time. In this paper, we propose the feature extraction method based on a filter bank to solve these problems. The proposed method consists of five steps. First, motor imagery EEG is divided by a using filter bank. Second, the regularized CSP (R-CSP) is applied to the divided EEG. Third, we select the features according to mutual information based on the individual feature algorithm. Fourth, parameter sets are selected for the ensemble. Finally, we classify using ensemble based on features. The brain-computer interface competition III data set IVa is used to evaluate the performance of the proposed method. The proposed method improves the mean classification accuracy by 12.34%, 11.57%, 9%, 4.95%, and 4.47% compared with CSP, SR-CSP, R-CSP, filter bank CSP (FBCSP), and SR-FBCSP. Compared with the filter bank R-CSP (β , γ), which is a parameter selection version of the proposed method, the classification accuracy is improved by 3.49%. In particular, the proposed method shows a large improvement in performance in the SSS.

Index Terms—Brain-computer interface (BCI), regularized common spatial pattern (R-CSP), filter bank, ensemble, small sample, motor imagery.

I. INTRODUCTION

BRAIN-COMPUTER INTERFACE (BCI) research using electroencephalograms (EEG) is actively in progress in

the fields of signal processing, cognitive science, medicine and rehabilitation [1]–[3]. These provide a way for people with physical disabilities to communicate with the outside world [4]. Technologies using BCI convert EEG to be generated using brain activity into a control signal so that an external device can be controlled without using muscles [5]. EEGs are widely used due to their many advantages, including economy, high temporal resolution, and simplicity [6]. The event-related desynchronization/event-related synchronization (ERD/ERS) pattern of the EEG that occurs during the motor imagery process is used as a very important feature in the BCI system [7]. However, the ERD/ERS pattern differs in signal patterns, frequency range, and location of the occurrence depending on the individual characteristics [8], [9]. In addition, EEG has a very low signal-to-noise ratio (SNR) and is non-stationary. Therefore, classification of EEGs is a difficult task [10], [11].

A common spatial pattern (CSP) was proposed to use the spatial features of ERD/ERS [12]. This algorithm consists of a spatial filtering technique, and it finds filters that maximize the variance for one class and minimize the variance for another class [13]. The CSP is very effective to classify motor imagery EEG, and therefore, it is widely used in the BCI system [14], [15].

However, the CSP highly depends on the frequency band. Therefore, it is necessary to select a wide frequency band or a frequency band that is suited to the individual characteristics, but this procedure is very inconvenient. In addition, the performance of the BCI system decreases significantly when the use of the frequency band is incorrect [16], [17]. The filter bank common spatial pattern (FBCSP) has been proposed to overcome such problems [18]. The FBCSP method divides the 4–40Hz frequency range of EEG into multiple smaller frequency bands, and it then applies the CSP for feature extraction. It thus overcomes the dependence of the frequency band of CSP, and it is effective in improving performance [19].

CSP is also highly dependent on sample-based covariance [20]. Therefore, CSP shows a low performance in the small-sample setting (SSS), and the regularized common spatial pattern (R-CSP) is proposed to overcome these problems [21]. This algorithm uses two parameters in a covariance estimation, and the R-CSP overcomes the performance limit of CSP in SSS by using two parameters that control the trade-off between the variance and the bias respectively. These studies

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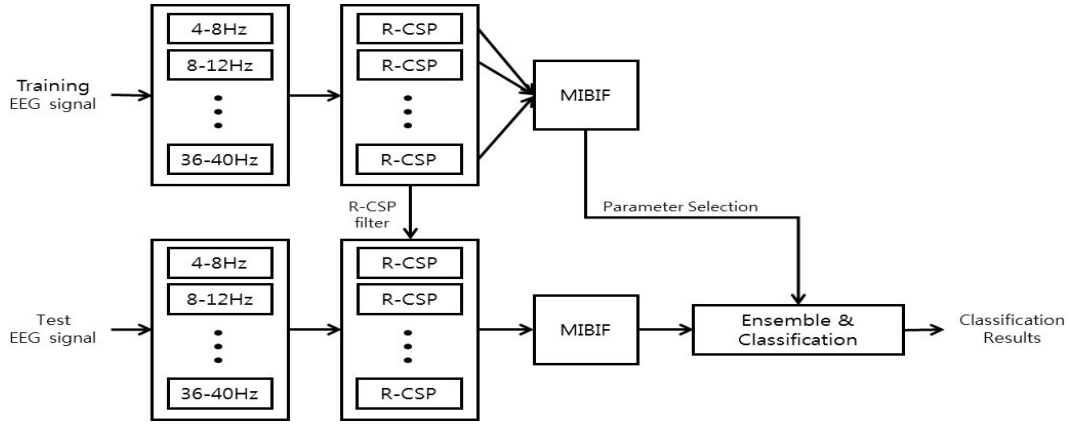


Fig. 1. The block diagram of the filter bank regularized the common spatial patterns ensemble method.

are intended to overcome dependence on the frequency band or covariance matrix estimation, respectively. These problems of CSP must be solved at the same time [22].

In this paper, a method is proposed to simultaneously solve the dependency on frequency band and the sample based covariance estimation. The proposed method is composed of five steps. First, EEG is divided by a filter bank that consists of a five-order butterworth bandpass filter. Second, R-CSP is applied for divided signals by the filter bank. Third, we select the feature pairs according to mutual information using the individual feature (MIBIF) algorithm. Fourth, parameter sets are selected for use with the ensemble. Finally, features are classified using the ensemble technique. In this paper, the performance of the algorithm is represented according to the classification accuracy.

The paper is organized as follows. Section II explains the proposed method, and Section III describes the BCI competition III dataset IVa as well as the experiment. Section IV compares and analyzes the results obtained through the experiment. Section V describes the conclusions regarding the method proposed in this paper as well as the direction for future studies.

II. METHODOLOGY

Our proposed method is based on the FBCSP framework, i.e., we use the same frequency bands as FBCSP (4-8Hz, 8-12Hz, ..., 36-40Hz) [18]. The block diagram of the proposed method is shown in Figure 1. The obtained EEG is filtered by a filter bank consisting of five-order butterworth bandpass filters containing nine different frequency bands [23]. The R-CSP is applied to filtered signals from each filter, and the motor imagery EEG is projected into the feature space using R-CSP. The k features are selected by the MIBIF from the extracted feature. Based on selected features, we apply an ensemble for classification. Each step is described in the subsections below.

A. Filter Bank Regularized Common Spatial Pattern (FBCSP)

The covariance that estimated in SSS leads to biased results. The estimated value that causes problems is difficult to trust,

and using the regularization technique controls the trade-off between the bias and the variance of the estimated value [21]. The regularization technique leads to a more “physically plausible” value for the sample-based estimated value [24]. Physiologically, the content of the discriminative information in the different frequency range differs for individuals [25]. Therefore, using the signal from different frequency ranges can be useful at the same time [26].

In this paper, we combine R-CSP with the filter bank structure to solve the problems for the frequency band dependence and the sample-based covariance estimation. The procedure of the filter bank regularizes the common spatial pattern (FBCSP) that was proposed in this paper as follows [21].

When the single trial EEG X_j filtered the j^{th} filter bank is a matrix of size $N \times S(Channel \times Sampling)$, the sample covariance matrix C_j for X_j is calculated as follows:

$$C = \frac{X_j X_j^T}{tr(X_j X_j^T)} \quad (1)$$

where $tr(\cdot)$ denotes the trace operator of the matrix, and the superscript T denotes the transpose of the matrix. The size of the calculated sample covariance matrix C_j is $N \times N$, and the sum of the sample covariance matrices for M trials of each class is calculated as follows:

$$C_{j,\omega} = \sum_{m=1}^M C_{(j,\omega,m)} \quad (2)$$

where M is the total trial number of each class, and m is the index of the trial ($m \leq M$). ω denotes the class index, and we consider only two classes in this paper, i.e., $\Omega = \{1, 2\}$. The generic covariance matrix $\hat{C}_{j,\omega}$ for regularization is calculated using the EEG collected for the other subjects as follows:

$$\hat{C}_{j,\omega} = \sum_{\hat{m}=1}^{\hat{M}} C_{(j,\omega,\hat{m})} \quad (3)$$

where $\hat{M}$ is the number of trials in class ω from the other subjects. Based on generic learning, $\hat{C}_{j,\omega}$ reduces the variance in calculating the regularized covariance matrix. In addition,

it provides more stable results [27]. The important point in Equation (3) is to calculate $\hat{C}_{j,\omega}$ by using EEG of subjects that excludes the interested one. We calculate the generic covariance matrix of all bands in the FBRCSF. $\hat{D}_{j,\omega}(\beta)$ is calculated to include not only the covariance matrix of a subject of interest, but also the covariance matrix of other subjects defined as follows:

$$\hat{D}_{j,\omega}(\beta) = \frac{(1 - \beta) \cdot C_{j,\omega} + \beta \cdot \hat{C}_{j,\omega}}{(1 - \beta) \cdot M + \beta \cdot \hat{M}} \quad (4)$$

where β controls the variance of the estimated covariance as the first regularization parameter. As the final regularization process, the regularized covariance matrix $\hat{\Sigma}_{j,\omega}(\beta, \gamma)$ for each class can be obtained as follows:

$$\hat{\Sigma}_{j,\omega}(\beta, \gamma) = (1 - \gamma) \cdot \hat{D}_{j,\omega}(\beta) + \frac{\gamma}{N} \text{tr}[\hat{D}_{j,\omega}(\beta)] \cdot I \quad (5)$$

where γ is the second regularized parameter, which functions as reduced large eigenvalues while increasing small eigenvalues, i.e., reduce bias caused by small samples.

The rest of the procedure is the same as the CSP method. The mixed covariance matrix $\hat{\Sigma}_j(\beta, \gamma)$ calculated using each class regularized covariance matrix. Then, it performs the eigenvalue decomposition as follows:

$$\hat{\Sigma}_j(\beta, \gamma) = \hat{\Sigma}_{j,1}(\beta, \gamma) + \hat{\Sigma}_{j,2}(\beta, \gamma) = U_j \lambda_j U_j^T \quad (6)$$

where U_j denotes the eigenvector, and λ_j denotes the eigenvalue. We can calculate the whitening transform matrix $Q_j = \sqrt{\lambda_j^{-1}} \cdot U_j^T$ using U_j and λ_j . The whitened spatial covariance matrix $S_{j,\omega}$ obtained as follows:

$$S_{j,\omega} = Q_j \cdot \hat{\Sigma}_{j,\omega}(\beta, \gamma) \cdot Q_j^T \quad (7)$$

$S_{j,1}$ and $S_{j,2}$ share the eigenvector, i.e.,

$$\begin{aligned} S_{j,1} &= B_j \Lambda_{j,1} B_j^T \\ S_{j,2} &= B_j \Lambda_{j,2} B_j^T \end{aligned} \quad (8)$$

where Λ is the eigenvalue corresponding the eigenvector, and $\Lambda_{j,1} + \Lambda_{j,2} = I$. The spatial filter W_j of the FBRCSF can be obtained using the eigenvector B_j as follows:

$$W_j = B_j^T Q_j \quad (9)$$

The signal X_j obtained from the filter bank is transformed by the corresponding spatial filter W_j as follows:

$$Z_j = W_j X_j \quad (10)$$

where Z_j denotes the signal transformed by a spatial filter W_j . To obtain the most discriminative feature from Z_j , we select d rows from each of the first and last rows. Therefore, $2d$ rows are selected. The feature vector $v_j = [v_j^1, v_j^2, \dots, v_j^{2d}]$ of FBRCSF for a single trial is calculated from the selected signal $\hat{Z}_j = [\hat{z}_j^1, \hat{z}_j^2, \dots, \hat{z}_j^{2d}]$ as follows:

$$v_j^k = \log \left(\frac{\text{var}(\hat{z}_j^k)}{\sum_{k=1}^{2d} \text{var}(\hat{z}_j^k)} \right) \quad (11)$$

where v_j^k is the feature value corresponding to signal $\hat{z}_j^k$ and k^{th} the feature vector component of v_j . We set $d = 3$ in this paper.

B. Mutual Information Based Individual Feature (MIBIF)

The classification error can be minimized by selecting the subset from the total feature set [28]. The mutual information based individual feature (MIBIF) algorithm is a feature selection method that is based on mutual information [29]. The MIBIF algorithm shows good performance in the filter bank-based method [18]. The producer of the MIBIF is divided into four stages. First, we set the selected feature vector set, the entire feature vector set, and the corresponding class label. Second, the mutual information of each feature is calculated. Third, the features are sorted in descending order according to mutual information. Fourth, the top k features and the corresponding feature pairs are selected. The features of CSP exist in pairs. Therefore, it should include the corresponding pair with the selected features [30].

According to previous studies [23] and [29], we apply the MIBIF algorithm to the obtained feature in the following order. The following procedure applies to all parameter sets, i.e., MIBIF is performed number of elements in $\beta \times$ number of elements in γ times. To simplify the explanation and understanding, the parameter index is not used.

- 1) Initialization: a set of selected feature vectors $S = \Phi$, a set total feature vectors $F = [f_1, f_2, \dots, f_{9 \times 2d}]$, $f = [v_1, v_2, \dots, v_M] \in \mathbb{R}^{M \times 1}$, and corresponding class label set Y are initialized.
- 2) Mutual information calculation: the mutual information $I(f_i; y)$ is calculated for each feature $f_1, f_2, \dots, f_{9 \times 2d}$.
- 3) Feature sorting: the set of total features F is arranged in descending order according to the mutual information values.
- 4) Features selection: the top k features are selected from the sorted feature vectors F , and the corresponding feature pairs are selected.

We set $k = 2$ in this paper. Therefore, the selected feature vector set S consists of two or four dimension feature vector. As the result of the MIBIF, we obtain number of elements in $\beta \times$ number of elements in γ selected feature vector sets.

C. Parameter Sets Selection

R-CSP can show lower performance than CSP during the actual test process when only one parameter set is selected through cross validation [21]. In addition, since only one parameter set is selected, the classification performance has a large deviation. Therefore, the performance of the BCI systems is unstable. In this paper, we select multiple parameter sets using five-fold cross validation and perform the ensemble. The parameter selection procedure is as follows.

Each class mean vector μ_ω is calculated in the training fold, i.e.,

$$\mu_\omega = \frac{1}{N_\omega} \sum_{s \in \omega} s \quad (12)$$

where s denotes the feature vector in class ω from the selected feature vector set S , N_ω denotes the number of feature vectors

belonging to class ω . The distance Dis_{ω} between each feature vector and each class mean vector is calculated as follows:

$$Dis_{\omega} = \|\mu_{\omega} - v\|_2 \quad (13)$$

where v denotes the feature vector in validation fold, and $\|\cdot\|_2$ denotes $L2$ - norm operator. As a result of Equation (13), we obtain Dis_1 and Dis_2 for one feature vector. Then compares the size of Dis_1 and Dis_2 and assigns the closest class. This procedure is applied to all folds. Then, we select the parameter sets that yielded the top 10% mean classification accuracy.

D. Ensemble Classification

Many studies have shown that the ensemble of predictive values is effective [31], [32]. Different sets of the regularization parameter sets derive the different features. In addition, since each parameter set is independent of each other, it is suitable for ensemble-based learning [21]. Therefore, we apply the ensemble method for each subject in this study. The obtained features from the selected parameter sets are used for the ensemble. For the ensemble, we use Equation (13) to compute the $L2$ -norm between each class for all test features. Then, to normalize the distance to [0, 1], we calculate it as follows [33]:

$$\overline{Dis}_{\omega,e} = \frac{Dis_{\omega,e} - \min(Dis)}{\max(Dis) - \min(Dis)} \quad (14)$$

where the subscript e denotes the ensemble index, $\min(Dis) = \min(\{Dis_{\omega,e}\}_{\omega=1}^{\Omega})$, and $\max(Dis) = \max(\{Dis_{\omega,e}\}_{\omega=1}^{\Omega})$. Finally, we can obtain the aggregated nearest-neighbor distance using a simple sum rule as follows [21]:

$$\overline{Dis}_{\omega} = \sum_{e=1}^E \overline{Dis}_{\omega,e} \quad (15)$$

For the test sample, we can assign a class label as follows:

$$\omega^* = \operatorname{argmin}(\overline{Dis}_{\omega}) \quad (16)$$

The Equation (16) can be solved very simple. We obtain the ensembled distance $\overline{Dis}_{\omega}$ for each class by equation (15). Thereafter, the class of the shortest distance among all distances is optimized class ω^* . The assigned label is used to compute the classification accuracy compared to the true label. Figure 2 shows the pseudo-code of proposed method in this paper.

III. DATA AND EXPERIMENTS

A. Data Description

In this paper, we use BCI competition III dataset IVa that is publicly available to measure the performance of the proposed algorithm [34]. This dataset is useful to evaluate the performance of the algorithms with a small amount of data, and it has been used in many studies [35]. The EEG signal was obtained from five healthy subjects ("aa", "al", "av", "aw", and "ay"). The subjects sat in a comfortable chair and performed motor imagery experiments. The EEG signal was recorded using 118 channels and 140 trials for each class.

TABLE I
BCI COMPETITION III DATASET VIA DESCRIPTION

Subject	Train	Test
aa	168	112
al	224	56
av	84	196
aw	56	224
ay	28	252

Input: A set of EEG trials for each class of all subjects, where class $\Omega = \{1, 2\}$. Regularization parameters $\beta, \gamma \in \{0, 0.1, \dots, 1\}$. Parameter d for obtaining the most discriminative features in transformed signal. Parameter k for selecting the features based on mutual information.

Output: The class label (ω^*) for test data.

FBRCS method:

Step 1. Feature extraction and Feature selection

for $i \leftarrow 1$, the number of subjects **do**

for $j \leftarrow 1$, the number of filter **do**

- Calculate the covariance $C_{j,\omega}$ for interested subject according to Eq. (1) and (2).
- Calculate the generic covariance $\hat{C}_{j,\omega}$ for other subjects according to Eq. (1) and (3).
- Obtain the R-CSP filters following Eq. (4), (5), (6), (7), and (9) for all β, γ .
- Obtain the transformed signal Z_j using Eq. (9)
- Extract the feature vectors using Eq. (10)

end for

- Apply MIBIF algorithm on total vectors

$$F = [f_1, f_2, \dots, f_{9 \times 2d}].$$

- Select the parameter sets for ensemble using 5-fold cross validation.

end for

Step 2. Ensemble

for $i \leftarrow 1$, the number of subjects **do**

for $j \leftarrow 1$, the number of selected parameter sets **do**

- Calculate the distance Dis_{ω} between mean feature vector from each class and feature vectors using Eq. (13).
- Normalize Dis_{ω} to [0, 1] to get $\overline{Dis}_{\omega,e}$ using Eq. (14).

end for

- Obtain the aggregated distance $\overline{Dis}_{\omega}$ using Eq. (15).

- **Output** Assign the class label for the test data using Eq. (16).

end for

Fig. 2. Pseudo-code implementation of the FBRCS method.

The classes consist of the right hand and foot, ie., the EEG signal was provided a total of 280 trials for each subject. The training and test data for each subject are shown in Table I.

The BCI competition III dataset IVa is available in 1,000Hz version and 100Hz version. In this experiment, we use a 100Hz version, and use 0.5 second to 2.5 seconds of 3.5 seconds of motor imagery EEG [36]. 18 EEG channels were used, as shown in Figure 3 (C1, C3, Cz, C2, C4, CFC3, CFC4, CFC5, CFC6, CCP3, CCP4, CCP5, CCP6, T7, T8, P3, Pz, and P4). The reference channel is Fpz. According to the homunculus theory, the selected channels contain the motor area and include the cognitive area needed to recognize the cue in the experiment [37].

B. Experiments

The computer specification used in this study is an Intel(R) Core(TM) i5-3470 CPU 3.2GHz with 12GB RAM with

TABLE II
CLASSIFICATION ACCURACY OF BCI COMPETITION III DATASET IVA

Subject	CSP	SR-CSP	R-CSP	FBCSP	SR-FBCSP	FBCSP(β, γ)	FBCSP
<i>aa</i>	75.89	72.32	75.89	88.39	86.61	90.17(0.1, 0.6)	91.07
<i>al</i>	91.07	92.85	91.07	94.64	92.85	92.85(0, 0)	94.64
<i>av</i>	56.63	60.71	60.2	61.22	67.85	69.38(0.2, 0)	75
<i>aw</i>	66.51	69.64	70.98	80.02	75	73.21(0.2, 0)	76.78
<i>ay</i>	79.36	77.77	88	82.14	86.50	88.09(0, 0.4)	93.65
mean	73.89	74.66	77.23	81.28	81.76	82.74	86.23

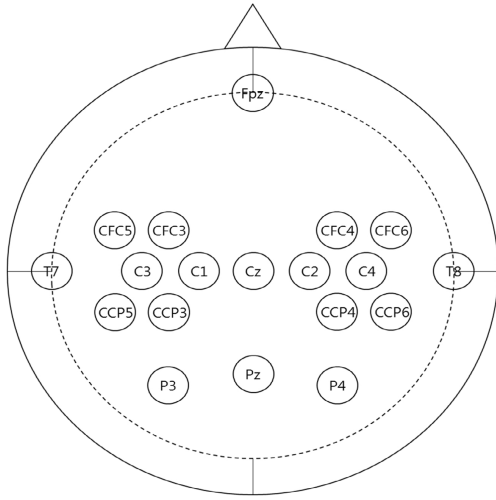


Fig. 3. Locations of the electrodes used.

Windows 7 Professional K 64bit. The proposed method and other methods are designed and tested using MATLAB R2016a.

The experiment consists of a total of five steps. First, pass the motor imagery EEG signal to the filter bank formed by dividing the frequency range of 4-40Hz into nine bands. The filter bank consists of a five-order butterworth bandpass filter. The range of each band is 4-8Hz, 8-12Hz, 36-40Hz. Second, apply R-CSP to the signal derived from the filter bank. We use three spatial filters from the first rows and the last rows, respectively, i.e., $m = 3$. The regularization parameters are $\beta, \gamma \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$. After R-CSP, the signal is projected in the feature space. Third, we select feature pairs by using the MIBIF. In this study, the MIBIF parameter k is set to 2. Fourth, we select the parameter sets for the ensemble. Finally, the label for each data is assigned using the ensemble.

The other methods are designed as follows; CSP and R-CSP methods use the 7-30Hz frequency bands that are generally used. SR-CSP and SR-FBCSP methods use the identity matrix as the covariance matrix estimator [38], [39]. In addition, the parameter setting of all methods is the same as the proposed one.

IV. RESULTS AND DISCUSSION

The performance of the FBCSP algorithm is verified by comparing it with the CSP, R-CSP, FBCSP, and FBCSP(β, γ) methods. The FBCSP(β, γ) method is a parameter selection version of the proposed method.

Table II shows the classification accuracy obtained using the BCI competition III dataset IVA from each method. We provide the selected parameter set in the FBCSP(β, γ) method. As shown in Table II, the proposed FBCSP method improves the mean classification accuracy by 12.34%, 11.57%, 9%, 4.95%, 4.47%, and 3.49% compared to the CSP, SR-CSP, R-CSP, FBCSP, SR-FBCSP, and FBCSP(β, γ) methods. The FBCSP yields a higher accuracy compared to the other methods for subjects “aa”, “al”, “av”, and “ay”. For subject “aw”, the FBCSP method obtained a 3.24% increase in accuracy than the FBCSP. In the ensemble process of the FBCSP, there are many parameter sets that interrupt with the performance. Therefore, a lower classification accuracy is obtained than when using the FBCSP method.

The R-CSP shows a good performance for subjects “av”, “aw”, and “ay” with SSS. Accordingly, R-CSP yielded better performance than CSP in subjects “av”, “aw”, and “ay”. Additionally, the same classification results as CSP were obtained for subjects “aa”, and “al”. The filter bank based methods solve the dependency on the frequency range and have shown an overall improvement in performance from a structural point of view. The FBCSP yielded a higher mean classification accuracy compared to the CSP, SR-CSP, and R-CSP, and it obtained a higher classification accuracy than CSP and SR-CSP for all subjects.

The proposed FBCSP method shows the advantages of R-CSP with a good performance in an SSS situation, and its advantages were obtained using a filter bank structure. Specially, FBCSP has a large performance improvement of 18.37%, 10.27%, and 14.29% for the subjects “av”, “aw”, and “ay” compared to CSP, respectively. These results show that the proposed method is very effective in SSS. On the other hand, FBCSP using the covariance estimates of CSP have a lower classification accuracy than the R-CSP for subject “ay” that provide the least amount of training data. The FBCSP has a dependency on the frequency range, but it shows a low performance in SSS. Therefore, FBCSP is suitable to overcome the performance degradation in SSS and the frequency band selection problem.

The FBCSP(β, γ) generally yielded results that include the advantages of both R-CSP and FBCSP. However, it has been difficult to find an optimal parameter set. For subjects “al” and “aw”, FBCSP(β, γ) exhibited a lower classification accuracy compared to the FBCSP. These results show that the parameter selection is wrong. On the other hands, since FBCSP uses an ensemble, it can be compensated by results

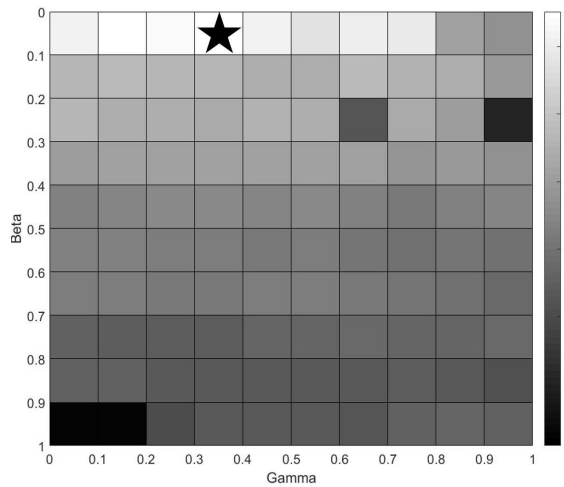


Fig. 4. The classification accuracy according to the parameter set for subject “aw”.

derived from other parameter sets even if the result derived from one parameter set is wrong. Therefore, more stable and high classification results is obtained.

Figure 4 shows the classification accuracy according to the parameter set for subject “aw”. In Figure 4, the x-axis represents the gamma value, and the y-axis represents the beta value. The brighter the color of the block, the higher the classification accuracy. The pentagram represents the set of parameters that yields the highest classification accuracy. Subject “aw” was significantly affected by the beta value, and it obtained generally good classification accuracy in $\beta = 0$, i.e., the subject “aw” is more effective in suppressing the occurrence of bias than the variance.

In the FBRCS(β, γ) method, subject “aw” selected parameter set $\beta = 0.2$ and $\gamma = 0$. Therefore, it obtained a lower classification accuracy than FBCSP. However, the highest classification accuracy of 82.59% was obtained in parameter set $\beta = 0$ and $\gamma = 0.3$. This result shows a difference of 9.38% in the classification rate with the selected parameter set. The FBRCS(β, γ) yields a classification accuracy of 50% in the worst case ($\beta = 1, \gamma = 0$ or $\beta = 1, \gamma = 0.1$), i.e., since the FBRCS(β, γ) selects only one parameter set, the variation in performance is very large.

Figure 5 shows the results of 10 times FBRCS(β, γ). In Figure 5, the x-axis represents the index of the experiment and the y-axis represents the classification accuracy. Subjects “aa” and “al” with a large number of data show stable classification performance. Subjects with large amounts of data are not significantly affected by the regularization process. Since there is a large quantity of data, the optimal parameter set is usually selected. On the other hand, subjects “av”, “aw”, and “ay” with SSS result in significantly different classification performance each time, i.e., the dependency on the generic dataset increase when the number of data is small. Therefore, the performance varies with sensitivity according to the selected parameter set. In the FBRCS(β, γ), the difference of the mean classification accuracy is of up to 7.63%, and this difference is mostly due to the performance differences in subjects “av”, “aw”, and “ay”, i.e., there is a larger

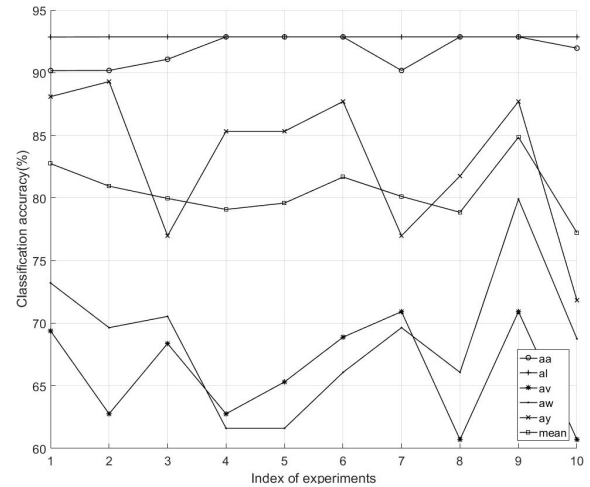


Fig. 5. Classification accuracy according to 10 times FBRCS(β, γ).

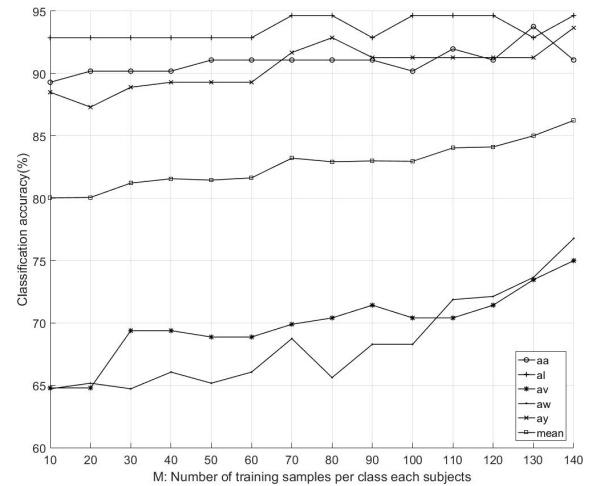


Fig. 6. Classification accuracy according to the number of M.

performance difference for the three subjects. The BCI systems that derive these results are difficult to trust.

Figure 6 shows the classification accuracy according to the number of data of the other subjects used for regularization. In Figure 6, the x-axis represents the number of each class of data for each subject used to calculate the generic covariance, and the y-axis represents the classification accuracy. A greater amount of data was used to calculate the generic covariance, and a better average classification accuracy was obtained. The mean classification accuracy difference between $M = 10$ and $M = 140$ is 6.2%. The more data there is, the greater change in covariance to a “physically plausible” value, and the classification accuracy improves. These results show that the proposed method follows the regularization theory.

As the number of data increases, the classification accuracy of the subjects “av” and “aw” greatly improves, showing a maximum performance difference of 11.2% and 12.05%, respectively. On the other hand, subject “ay” yielded a classification accuracy of 88.49% in $M = 10$. This result shows the best classification accuracy compared to the other methods. Subject “ay” is affected by the generic covariance because the number of training data is very small. Therefore, subject “ay” has a sufficiently good performance even if M is small.

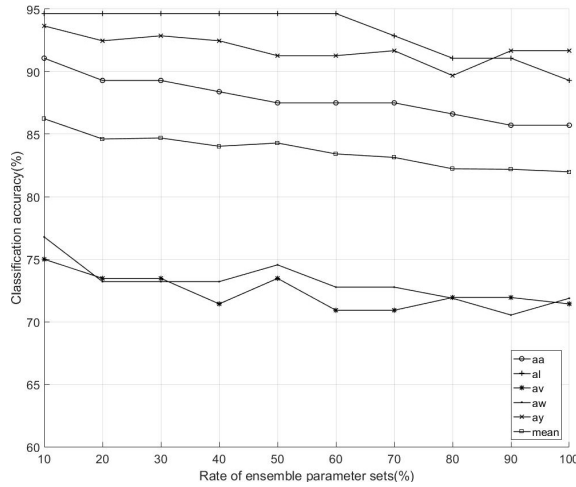


Fig. 7. Classification accuracy according to the number of parameters for the ensemble.

In addition, even though “ay” has fewer training data than “av” and “aw”, so a better performance in all methods means that the collected data for “ay” is very satisfactory.

The FBRCSP yielded a higher classification accuracy than CSP and R-CSP in $M = 10$. To obtain better performance for all subjects. These results show that the amount of data to construct a generic covariance yields at least an effective performance. At the same time, a higher classification accuracy was obtained for subjects “aa” and “al” with a large training data through the filter bank structure.

Figure 7 shows the classification accuracy according to the number of parameters for the ensemble. The ratio of the parameters was divided into 10% units, and experiments were performed from 10% to 100%. In Figure 7, the x-axis represents the ratio of the parameters for the ensemble, and the y-axis represents the classification accuracy. As the number of parameters for the ensemble increases, the average classification accuracy decreases. This tendency occurs because among all parameter sets, an increasing number of parameter sets interfered with the classification. At an ensemble parameter ratio of 10%, the mean classification accuracy is 81.99%, which is better than the CSP, R-CSP and FBCSP. In addition, the mean difference in the classification accuracy between it and 100% is of 4.24%, i.e., the difference in the mean classification accuracy according to the number of ensemble parameters is small. The results of the classification are very stable, even if the parameter sets that interfere with the classification are included due to using the simple rule of the sum of all distance for the ensemble.

Figure 8 shows the training time and test time for each method. In Figure 8, the x-axis represents methods, and the left y-axis and the right y-axis represent training time and test time, respectively. The CSP and the SR-CSP have low performance but require very short training time. The FBCSP and the SR-FBCSP require similar time. CSP and SR-CSP based methods do not require a cross validation procedure for parameter selection. Therefore, training time is short. However, methods based on R-CSP require cross-validation, i.e., a lot of training time is needed compared to other methods.

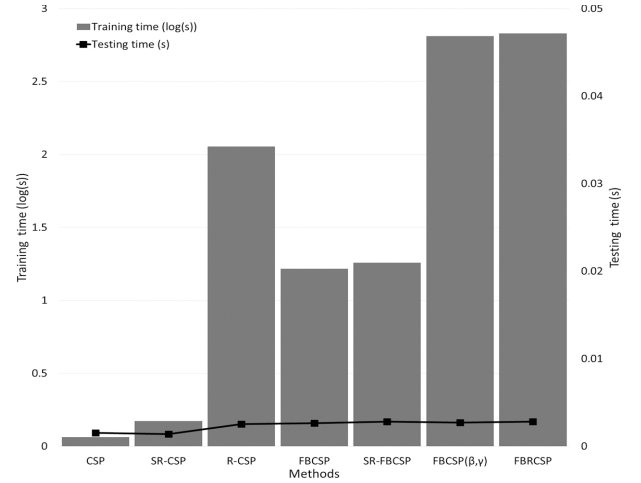


Fig. 8. The required time for training and test.

On the other hand, the test time is very short in all methods. The required test time for one trial is less than 0.01 seconds. Therefore, if training is performed off-line and then test online afterwards, it can overcome the time-consuming issues.

V. CONCLUSION

In this paper, we proposed a method to overcome two problems of CSP. The CSP has a sample-based covariance matrix estimation problem in SSS, and it is dependent on the frequency bands. To solve these problems, we combine R-CSP and a filter bank structure, and also perform an ensemble. This method is an extended version of FBCSP and R-CSP. The proposed method can overcome the dependency on the frequency bands using the filter-bank structure, and it can overcome the performance degradation problem in the SSS situation using R-CSP. The proposed method offers both advantages of the filter bank structure and advantages of R-CSP. In addition, we have solved the disadvantages of the parameter set selection method through the ensemble.

The proposed method showed a better classification performance for motor imagery EEG than CSP, SR-CSP, R-CSP, FBCSP, SR-FBCSP, and FBRCSP(β, γ). In particular, the proposed method showed a greater performance in SSS, and we confirmed that controlling the trade-off between the bias and the variance for the signal derived from the filter bank has a great effect on the performance of the BCI system. Thus, this ensemble technique helps derive stable classification results.

In conclusion, the method proposed in this paper can simultaneously solve the dependency on the frequency bands, and the performance degradation in the SSS as well as problems due to parameter selection. There are many problems surrounding BCI. However, there is a lack of research to combine and solve multiple problems. Hereafter, studies dealing with further BCI issues as well as issues dealt with in this paper should be continued.

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